

Project Design Phase

Solution Architecture

Date	02 JULY 2026
Team ID	
Project Name	India's Agricultural Crop Production Analysis
Maximum Marks	4 Marks

Solution Architecture:

- The solution begins by collecting India's agricultural cultivation dataset (1997–2021) from CSV/Excel files.
- The collected data is cleaned and preprocessed by removing null values, correcting data types, and validating the dataset.
- The processed data is imported into Tableau Desktop for analysis and visualization.
- Interactive worksheets, dashboards, KPI cards, maps, and Tableau Stories are created to present agricultural insights.
- Users can analyze data using filters such as **State, Crop, Season, and Year**.
- The dashboard provides state-wise, crop-wise, seasonal, and yearly production analysis through interactive visualizations.
- The solution supports multiple stakeholders, including **farmers, government policy analysts, agribusiness investors, researchers, and students**.
- The final outcome enables data-driven decision-making, improves agricultural planning, supports policy formulation, and identifies profitable investment opportunities.

Example - Solution Architecture Diagram:

